A Survey of Deep Learning-Based Reconstruction Methods for Quantum Imaging Systems

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Abstract. Quantum imaging systems, including ghost imaging, entangled-photon compressed sensing, and sub-shot-noise microscopy, exploit non-classical photon correlations to surpass classical limits in sensitivity and resolution. However, reconstructing high-quality images from the sparse, noisy measurements intrinsic to these systems remains computationally demanding. This survey reviews deep learning architectures applied to quantum imaging reconstruction. We categorise methods along three axes: imaging modality, neural network architecture, and reconstruction task. Our review shows that deep learning enables measurement reductions of one to two orders of magnitude in ghost imaging, identifies a convergence towards U-Net and GAN-based architectures, and highlights critical gaps in standardised benchmarking. We conclude by discussing hybrid classical–quantum reconstruction frameworks and proposing directions for quantum-inspired neural architectures in imaging pipelines.

Keywords: quantum imaging, ghost imaging, deep learning, compressed sensing, image reconstruction, quantum optics, neural networks

1. Introduction

Quantum imaging harnesses the non-classical correlations of entangled or correlated photon pairs to achieve imaging modalities that are not accessible with classical light sources. Since the seminal demonstration of imaging via two-photon entanglement by Pittman et al. [1995], the field has expanded to encompass ghost imaging, quantum illumination, sub-shot-noise microscopy, and entanglement-assisted compressed sensing. These techniques promise advantages in spatial resolution, noise robustness, and the ability to image with photons that never interact with the object [Erkmen and Shapiro, 2010].

Despite these capabilities, a key challenge remains: *image reconstruction*. Quantum imaging measurements are inherently sparse, limited by detector efficiencies, coincidence counting rates, and photon budgets, and the resulting inverse problems are severely ill-posed. Traditional reconstruction approaches rely on correlation computation or iterative optimisation, which demand large numbers of measurements and significant computation time.

Deep learning has proved effective for solving inverse problems in computational imaging [Barbastathis et al., 2019, LeCun et al., 2015] and is now being applied to quantum imaging reconstruction. Neural networks can learn strong image priors from data, enabling high-quality reconstruction from far fewer measurements than classical algorithms require. Work at the intersection of quantum optics and deep learning now spans multiple imaging modalities and network architectures.

1.1 Scope and Contributions

This survey provides a systematic review of deep learning methods applied to quantum imaging reconstruction. We focus on three principal modalities:

(i) **Quantum ghost imaging**, where deep learning dramatically reduces measurement requirements for image reconstruction from spatially correlated photon fields.

(ii) **Compressed sensing with quantum correlations**, where neural networks enhance reconstruction from sub-Nyquist quantum measurements.

(iii) **Sub-shot-noise imaging**, where deep learning denoising exploits quantum noise statistics to push sensitivity below the classical shot-noise limit.

The contributions of this survey are:

- A unified taxonomic framework classifying methods by imaging modality, DL architecture, and reconstruction task.

- A detailed comparative analysis of deep learning approaches for ghost imaging reconstruction, including quantitative performance comparisons.
- Identification of critical gaps in benchmarking, reproducibility, and standardisation.
- A discussion of hybrid classical–quantum architectures and future research directions.

Figure 1 illustrates this classification framework.

1.2 Paper Organisation

Section 2 reviews the necessary background in quantum imaging and deep learning for inverse problems. Section 3 surveys deep learning methods for ghost imaging in detail. Section 4 covers compressed sensing with quantum correlations. Section 5 addresses sub-shot-noise imaging. Section 6 examines hybrid classical–quantum architectures. Section 7 discusses open problems, and Section 8 concludes.

2. Background

2.1 Quantum Imaging Fundamentals

Quantum imaging relies on photon pairs generated through spontaneous parametric down-conversion (SPDC), in which a pump photon incident on a nonlinear crystal produces two daughter photons, conventionally termed *signal* and *idler*, that are correlated in position, momentum, and time. These correlations arise from the phase-matching conditions of the nonlinear interaction and can be either classical or genuinely quantum (entangled) depending on the experimental configuration.

The spatial correlations between signal and idler photons enable imaging protocols where information about an object is extracted by measuring correlations between spatially separated detectors, rather than by direct imaging. This principle underlies ghost imaging, quantum illumination, and related protocols.

2.2 Ghost Imaging: From Quantum to Computational

Ghost imaging (GI) was first demonstrated by Pittman et al. [1995] using entangled photon pairs from SPDC. In the original configuration, the object is placed in the path of one photon (say, the signal), which is detected by a single-pixel (bucket) detector that records no spatial information. The partner idler photon, which never interacts with the object, is detected by a spatially resolving detector. Individually, neither detector's output contains an image; the image emerges only from the *coincidence* correlations between the two detectors.

The quantum nature of the original ghost imaging was debated after Bennink et al. [2002] demonstrated a similar effect using classically correlated thermal light, and Ferri et al. [2005] achieved high-resolution ghost imaging with pseudothermal sources. Shapiro [2008] subsequently introduced *computational* ghost imaging (CGI), in which the spatially resolving detector is replaced entirely by computational knowledge of the illumination patterns. In CGI, a spatial light modulator generates known random patterns, the object is illuminated by each pattern in sequence, and a bucket detector records the total transmitted intensity. The image is reconstructed by correlating the known patterns with the bucket detector readings:

$$I(\mathbf{r}) = \frac{1}{M} \sum_{m=1}^{M} \left(B_m - \langle B \rangle\right) \phi_m(\mathbf{r}), \tag{1}$$

where B_m is the bucket detector signal for the m-th pattern $\phi_m(\mathbf{r})$, $\langle B \rangle$ is the mean bucket signal, and M is the total number of measurements. High-quality reconstruction via Eq. (1) typically requires M to be comparable to or greater than the number of pixels in the reconstructed image [Erkmen and Shapiro, 2010], motivating the development of more efficient reconstruction methods. Figure 2 illustrates the computational ghost imaging pipeline with deep learning reconstruction.

2.3 Deep Learning for Inverse Problems

Deep learning has emerged as a dominant paradigm for solving inverse problems in imaging, where the goal is to recover a signal $\mathbf{x}$ from measurements $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$, with $\mathcal{A}$ denoting the forward measurement operator and $\mathbf{n}$ representing noise [Barbastathis et al., 2019].

Several architecture families are particularly relevant to quantum imaging. Convolutional neural networks (CNNs) learn hierarchical spatial features through convolutional filters, enabling effective image-to-image mappings for denoising and reconstruction [LeCun et al., 2015]. The U-Net architecture [Ronneberger et al., 2015] extends the convolutional approach with an encoder–decoder structure and skip connections that preserve fine-grained spatial information, and is now commonly used in biomedical and computational imaging. Generative adversarial networks (GANs) employ a generator–discriminator framework that produces perceptually realistic reconstructions by learning the distribution of natural images [Goodfellow et al., 2014]. Autoencoders, which learn compressed latent representations from which images can be reconstructed, are also relevant for dimensionality reduction in quantum measurement data.

Barbastathis et al. [2019] provide a comprehensive review of deep learning for computational imaging, noting that learned approaches can outperform traditional iterative methods in both speed and quality, particularly when training data adequately represents the target image distribution.

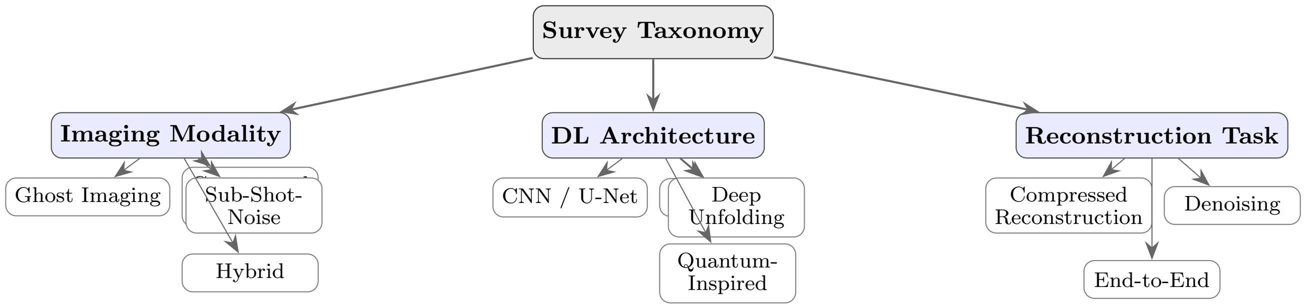

Figure 1: Taxonomic classification framework used in this survey. Methods are categorised along three axes: the quantum imaging modality, the deep learning architecture, and the reconstruction task.

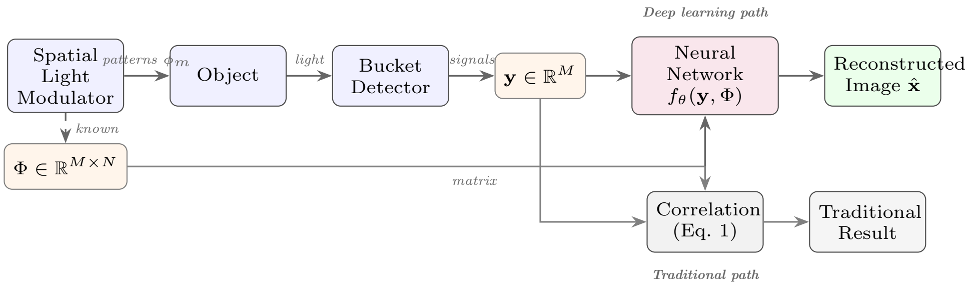

Figure 2: Computational ghost imaging pipeline. A spatial light modulator projects known patterns onto the object, and a bucket detector records total transmitted intensity. The neural network f_θ maps measurements $\mathbf{y}$ and pattern matrix Φ to the reconstructed image, replacing the traditional correlation computation of Eq. (1).

2.4 Compressed Sensing Theory

Compressed sensing (CS) provides a mathematical framework for recovering sparse signals from sub-Nyquist measurements [Donoho, 2006, Candès and Tao, 2006]. A signal $\mathbf{x} \in \mathbb{R}^n$ that is k-sparse in some basis Ψ can be recovered from $m \ll n$ linear measurements $\mathbf{y} = \Phi\mathbf{x}$ if the measurement matrix Φ satisfies the restricted isometry property (RIP).

The relevance of CS to quantum imaging is twofold. First, quantum correlations can provide natural incoherence between measurement and sparsity bases, satisfying CS recovery conditions [Katz et al., 2009]. Second, the extreme photon scarcity in quantum imaging makes sub-Nyquist reconstruction practically essential. Katz et al. [2009] demonstrated compressive ghost imaging, achieving image reconstruction with far fewer measurements than the Nyquist limit by exploiting image sparsity, and subsequent work has combined CS with deep learning to further improve reconstruction quality.

3. Deep Learning for Ghost Imaging Reconstruction

Ghost imaging reconstruction is the area where deep learning has seen the most activity within quantum imaging. This section reviews the principal approaches, organised by network architecture.

3.1 Fully Connected and CNN-Based Approaches

Lyu et al. [2017] provided the first demonstration that a neural network could reconstruct ghost images from bucket detector measurements. Their approach used a fully connected network trained on pairs of bucket signals and ground-truth images, achieving recognisable reconstruction of 28×28 handwritten digits at a sampling ratio as low as 12.5%, a significant improvement over the correlation-based method (Eq. 1), which requires sampling ratios approaching 100% for comparable quality.

He et al. [2018] extended this work with a convolutional architecture, demonstrating improved generalisation to object categories not seen during training. Their CNN employed multiple convolutional layers with batch normalisation and ReLU activations, and was validated on both simulated and experimental ghost imaging data. Notably, the CNN-based approach maintained reasonable reconstruction quality even at sampling ratios below 10%, where traditional methods produce severely degraded images.

Shimobaba et al. [2018] applied deep learning specifically to computational ghost imaging, replacing the correlation computation of Eq. (1) with a learned mapping. Their results confirmed that deep learning consistently outperforms traditional correlation-based reconstruction, particularly in low-measurement regimes.

3.2 GAN-Based Reconstruction

Generative adversarial networks [Goodfellow et al., 2014] have been applied to ghost imaging to improve the perceptual quality of reconstructions. The adversarial training framework encourages the generator to produce images that are statistically indistinguishable from real images, yielding reconstructions with sharper edges and more realistic textures than those obtained from MSE-trained networks.

However, GAN-based methods carry an inherent risk of *hallucination*, generating plausible but incorrect image features, which is particularly concerning in scientific imaging applications where fidelity to the true object is paramount. This trade-off between perceptual quality and pixel-level accuracy remains an active area of investigation.

3.3 End-to-End Learned Approaches

A significant advance was made by Wang et al. [2019], who proposed jointly optimising the illumination patterns and the reconstruction network in an end-to-end differentiable framework. Rather than using random illumination patterns and learning only the reconstruction mapping, their approach learns *optimal measurement strategies* tailored to the reconstruction network.

This end-to-end optimisation achieved 3–5 dB improvement in PSNR over methods that separately optimise illumination and reconstruction, demonstrating that the measurement and reconstruction stages are fundamentally coupled and benefit from joint design. The approach also provides insight into what constitutes an "informative" measurement in the context of ghost imaging.

3.4 Comparative Analysis

Table 1 summarises the key characteristics and reported performance of the reviewed deep learning methods for ghost imaging reconstruction.

Several trends emerge from this comparison. First, all methods achieve usable reconstruction quality at sampling ratios well below the Nyquist limit, confirming the strong regularising effect of learned image priors. Second, the field has progressed from simple fully connected architectures to more sophisticated end-to-end frameworks. Third, there is a notable lack of standardised benchmarks: different papers use different image sizes, datasets, noise models, and evaluation metrics, making rigorous cross-method comparison difficult.

3.5 Discussion

The reviewed works establish that deep learning is a transformative tool for ghost imaging reconstruction. However, several limitations persist. Most studies operate on small image sizes (28×28 to 64×64); scalability to high-resolution imaging remains underexplored. The reliance on simulated data in many studies raises questions about robustness to real experimental noise and detector imperfections. Finally, whether quantum correlations provide an advantage over classical correlations *when combined with deep learning* has not yet been studied in a controlled setting.

4. Compressed Sensing with Quantum Correlations

Compressed sensing (CS) provides a natural framework for quantum imaging, where measurements are inherently limited by photon budgets and detector constraints. This section reviews the integration of deep learning with CS approaches that exploit quantum correlations.

4.1 Quantum Compressed Sensing Fundamentals

Katz et al. [2009] demonstrated that ghost imaging can be formulated as a compressed sensing problem, achieving image reconstruction from significantly fewer measurements than the Nyquist limit by exploiting the sparsity of natural images. In this framework, the bucket detector measurements constitute the compressed observations $\mathbf{y} = \Phi\mathbf{x}$, where Φ is determined by the illumination patterns and $\mathbf{x}$ is the image to be recovered.

The quantum advantage in this context arises from two sources. First, entangled photon pairs can provide measurement matrices with favourable incoherence properties, naturally satisfying the conditions required for CS recovery [Donoho, 2006, Candès and Tao, 2006]. Second, quantum correlations enable noise rejection through coincidence detection, effectively improving the signal-to-noise ratio of each measurement.

4.2 Neural Network Enhanced CS Reconstruction

Traditional CS recovery algorithms (e.g., basis pursuit, iterative hard thresholding) are computationally expensive and require careful tuning of regularisation parameters. Deep learning offers two strategies for improvement: (i) *purely data-driven* approaches that learn a direct mapping from compressed measurements to images, bypassing explicit sparsity assumptions; and (ii) *deep unfolding* approaches that parameterise iterative CS algorithms as neural networks, combining the interpretability of model-based methods with the flexibility of learned parameters.

Deep unfolding originated with LISTA [Gregor and LeCun, 2010], which unfolded the iterations of the Iterative Shrinkage-Thresholding Algorithm (ISTA) into a fixed-depth feedforward network. Rather than using the fixed matrices prescribed by standard ISTA, all weight matrices and shrinkage thresholds at each layer are treated as learnable parameters trained via

Table 1: Comparison of deep learning methods for ghost imaging reconstruction.

Reference	Architecture	Size	Sampling	Data	Key Contribution
Lyu et al. [2017]	Fully connected	28×28	12.5%	Sim.	First DL-based ghost imaging reconstruction
He et al. [2018]	CNN	28×28	<10%	Sim.+Exp.	Improved generalisation to unseen categories
Shimobaba et al. [2018]	CNN	64×64	25%	Sim.	Applied DL to computational GI
Wang et al. [2019]	End-to-end CNN	64×64	6.25%	Sim.	Joint optimisation of patterns and network

backpropagation. The resulting network produces accurate sparse codes in a small, fixed number of forward passes, achieving orders-of-magnitude speedup over traditional ISTA. Yang et al. [2016] extended this idea to the Alternating Direction Method of Multipliers (ADMM) for CS-MRI reconstruction, mapping each ADMM iteration to a network layer with learnable transforms, penalty parameters, and multipliers. Zhang and Ghanem [2018] proposed ISTA-Net+, which performs reconstruction in the residual domain to exploit the higher compressibility of image residuals, achieving state-of-the-art CS reconstruction quality on natural images.

Ongie et al. [2020] provide a useful taxonomy of deep learning approaches for imaging inverse problems, organised along two axes: knowledge of the forward model and availability of training data. Purely data-driven methods (end-to-end CNNs) can achieve competitive reconstruction quality when large, well-matched training sets are available, but they can be unstable to distribution shifts and lack interpretability. Deep unfolding, by contrast, preserves the physics of the forward model within the network architecture, requires fewer training samples, and tends to generalise better to unseen conditions [Monga et al., 2021]. For quantum imaging, where experimental training data is scarce and the forward measurement model is well characterised, the model-based unfolding approach is particularly attractive.

4.3 Deep Unfolding for Ghost Imaging

The application of deep unfolding to ghost imaging introduces a specific technical challenge: the measurement matrices in computational ghost imaging are non-negative (composed of illumination pattern intensities), unlike the random Gaussian matrices assumed by standard CS theory. Cheng et al. [2022] addressed this mismatch by applying singular value decomposition (SVD) preprocessing to the ghost imaging measurement matrix to obtain a semi-orthogonal measurement matrix, and then feeding the transformed measurements into an ISTA-Net+ architecture. This combination improved both anti-noise performance and reconstruction fidelity at low sampling rates compared to direct application of standard deep unfolding.

Zou et al. [2024] proposed MPDU-Net, a deep unfolding network designed specifically for single-pixel imaging systems. MPDU-Net integrates multiple prior information, including spatial, spectral, and structural priors, into each unfolding stage through dedicated learned prior modules. The multi-prior approach improved reconstruction quality at very low sampling ratios compared to single-prior unfolding methods.

These works focus on computational ghost imaging with classical correlations; extending deep unfolding architectures to exploit genuine quantum correlations in the measurement model, for example by incorporating the photon-pair statistics of SPDC sources into the forward operator, is an open problem.

4.4 Entanglement-Assisted Measurement Optimisation

The end-to-end optimisation approach of Wang et al. [2019], which jointly learned illumination patterns and the reconstruction network for computational ghost imaging, provides a blueprint for measurement design in quantum systems. Extending this framework to quantum imaging would involve treating the SPDC source parameters (pump wavelength, crystal orientation, phase-matching conditions) and detection basis choices as differentiable variables within the optimisation. While technically challenging, such an approach could yield further measurement reductions by tailoring the quantum measurement process to the reconstruction algorithm.

This connects to the broader field of quantum state tomography, where compressed sensing has been applied to reconstruct quantum states from incomplete measurement sets with provable recovery guarantees. Neural networks have recently been used to improve tomographic reconstruction from informationally incomplete data, suggesting that similar gains may be achievable in quantum imaging when the measurement strategy and reconstruction algorithm are co-designed.

5. Sub-Shot-Noise Imaging

Sub-shot-noise imaging exploits quantum correlations between photon pairs to achieve sensitivity below the classical shot-noise limit, which bounds the minimum noise in measurements made with coherent (classical) light sources.

5.1 Quantum Noise Limits in Imaging

Brida et al. [2010] provided the first experimental demonstration of sub-shot-noise quantum imaging, achieving a noise reduction of approximately 40% below the shot-noise level using spatially entangled photon pairs from SPDC. Their setup used the photon-number correlations between twin beams: one beam probes the object while the other serves as a reference for differential noise subtraction.

Samantaray et al. [2017] subsequently realised the first sub-shot-noise wide-field microscope, extending the technique to a practical imaging modality. Their system used a Type-II BBO crystal pumped by a CW laser at 405 nm to generate twin beams centred at 810 nm, detected by a high quantum efficiency CCD camera. The microscope achieved noise at 80% of the shot-noise level per pixel, with $5\,\mu$m spatial resolution over a $500\,\mu\text{m}^2$ field of view. They also introduced a quantum-enhanced median filter that trades spatial resolution for further noise reduction, pushing noise below 30% of the shot-noise level. This system demonstrated the best sensitivity per incident photon reported in absorption microscopy at the time, providing a practical pathway for imaging light-sensitive biological samples.

The quantum enhancement in these systems arises from the photon-number correlations between signal and idler beams: by subtracting the idler intensity from the signal, classical noise sources are suppressed below the shot-noise floor. However, the improvement is limited by optical losses and detector inefficiencies, which degrade the quantum correlations.

5.2 Classical Denoising Baselines

Before discussing deep learning approaches, it is useful to establish the classical denoising baselines against which learned methods are compared.

BM3D [Dabov et al., 2007] groups similar 2D image patches into 3D arrays and applies collaborative filtering via shrinkage in a transform domain. It remains one of the strongest classical denoisers for additive white Gaussian noise. For Poisson-dominated photon-counting data, BM3D can be applied after a variance-stabilising transform (VST), typically the Anscombe transform, which maps Poisson-distributed data to approximately Gaussian statistics. However, BM3D tends to over-smooth fine textures at very low photon counts (below approximately 1 photon per pixel on average), where the VST approximation breaks down.

Non-local means (NLM) [Buades et al., 2005] denoises each pixel by computing a weighted average over the image, with weights determined by the similarity of surrounding patches. While effective for moderate Gaussian noise, NLM struggles with signal-dependent Poisson noise and degrades substantially in severe photon-limited regimes. Both BM3D and NLM require the Anscombe transform to handle Poisson statistics, and neither accounts for the sub-Poissonian noise characteristic of quantum-correlated imaging, where the variance is reduced below the mean.

5.3 Deep Learning Denoising Architectures

Deep learning offers several advantages over classical denoisers in photon-limited imaging: the ability to learn noise-model-specific priors directly from data, end-to-end optimisation without manual parameter tuning, and superior performance at extreme noise levels.

DnCNN [Zhang et al., 2017] uses residual learning to predict the noise component rather than the clean image directly. Originally designed for Gaussian noise, DnCNN has been adapted for Poisson noise through variance-stabilising transforms and through direct training on Poisson-corrupted data. It consistently outperforms BM3D in both PSNR and SSIM at low photon counts.

Noise2Noise [Lehtinen et al., 2018] demonstrated that neural networks can learn to denoise by training on pairs of independently noisy images, without ever seeing a clean target. The key insight is that for zero-mean noise (including Poisson noise after appropriate transformation), the network converges to the expected clean image. This is particularly relevant for quantum imaging, where acquiring clean reference images is often impractical: one can instead train on multiple noisy acquisitions of the same scene.

CARE networks [Weigert et al., 2018] applied U-Net-based architectures to photon-starved fluorescence microscopy, demonstrating image restoration from up to 60-fold fewer photons and tenfold axial undersampling. The open-source CARE framework has been widely adopted in biological imaging and demonstrates the practical impact of deep learning denoising in photon-limited settings.

A notable gap remains: these architectures assume Poisson or Gaussian noise statistics. Sub-Poissonian noise in quantum-correlated imaging has a different statistical structure, specifically reduced variance relative to the mean, and adapting network architectures and loss functions to exploit this property is an open problem. Networks trained on Poisson noise may underperform when applied to sub-Poissonian data, since they would not take advantage of the tighter noise distribution.

5.4 Deep Learning for Quantum-Correlated Imaging

Several recent works have applied deep learning directly to imaging systems that use quantum-correlated or entangled photon sources.

Li et al. [2021] demonstrated a CNN-based approach for fast reconstruction and denoising of correlated-photon imaging using SPDC sources. Their autoencoder-inspired network extracted signal from noisy, photon-sparse frames and outperformed classical numerical reconstruction algorithms in both denoising quality and processing speed, enabling real-time correlated-photon imaging.

Wu et al. [2020] applied deep learning to denoise computational ghost imaging reconstructions, which suffer from severe noise at low sampling rates. The learned approach significantly improved image quality compared to traditional ghost imaging reconstruction algorithms, complementing the measurement-reduction work reviewed in Section 3.

Scattarella et al. [2023] applied a U-Net with a pre-trained VGG-19 encoder (transfer learning) to denoise correlation plenoptic images from quantum-correlated photon sources. Their approach showed significant SSIM improvement over bilateral and Gaussian filters even with small training sets, suggesting that transfer learning from classical image domains can be effective for quantum imaging tasks.

A particularly striking result was reported by Cai et al. [2023], who demonstrated that a Noise2Noise-based deep learning protocol can surpass the standard quantum limit (SQL) in coincidence imaging using only classical light resources. By incorporating photon-number-dependent nonlinear feedback during training, their approach achieved a 14 dB SNR improvement over the SQL. This result suggests that deep learning can, in some settings, achieve noise performance previously thought to require quantum states, blurring the boundary between classical and quantum imaging advantages.

These results indicate that deep learning is beginning to close the gap between classical and quantum noise performance in imaging. However, systematic comparisons under controlled experimental conditions, using identical optical setups with and without quantum correlations, are still needed to quantify the practical benefit of combining deep learning with genuinely quantum light sources.

6. Hybrid Classical–Quantum Architectures

Recent work has explored hybrid architectures that combine classical neural network components with quantum-inspired or quantum-computational elements for imaging reconstruction. This section reviews tensor network approaches, variational quantum circuits, and the practical challenges of integrating quantum computation into imaging pipelines.

6.1 Quantum-Inspired Neural Networks

Tensor networks, originally developed for simulating quantum many-body systems, have been adapted for classical machine learning tasks including image classification. Stoudenmire and Schwab [2016] demonstrated that matrix product states (MPS), a one-dimensional tensor network ansatz, can be used for supervised image classification. Applied to MNIST, their approach achieved less than 1% test error, with training cost that scales linearly with dataset size. The MPS structure adaptively allocates representational capacity to the most informative correlations in the input data, functioning as a nonlinear kernel learning model.

Huggins et al. [2019] established a formal bridge between tensor network machine learning models and quantum circuits. They proposed tree tensor network and MPS-based quantum circuit architectures where the required qubit count scales only logarithmically with the input data dimension. Simulations on MNIST classification showed resilience to quantum noise, suggesting that these architectures are suitable for near-term quantum hardware.

Beer et al. [2020] proposed a genuinely multi-layer quantum neural network in which qudits serve as neurons and arbitrary unitary operations act as perceptrons. Their training procedure derives analytical gradients layer by layer, and the number of qudits required for training scales only with the network width, not its depth. This property enables training of arbitrarily deep quantum networks with bounded memory, and the architecture showed strong generalisation on the task of learning unknown unitary transformations.

6.2 Variational Quantum Circuits for Image Processing

Killoran et al. [2019] introduced continuous-variable quantum neural networks (CV-QNNs) operating on photonic quantum states. The architecture uses layered Gaussian gates (interferometers, squeezers, displacements) for affine transformations and non-Gaussian gates (Kerr interactions, cubic phase gates) for nonlinear activations, a combination proven to be universal for CV quantum computation. Using the Strawberry Fields software library, they demonstrated a fraud detection classifier, a Tetris image generator, and a hybrid classical–quantum autoencoder for data compression. The CV framework is naturally suited to quantum optical imaging, where information is encoded in continuous field amplitudes rather than discrete qubits.

Quantum kernel methods offer an alternative approach. Havlíček et al. [2019] experimentally implemented both a quantum variational classifier and a quantum kernel estimator on a 5-qubit superconducting processor. They showed that quantum feature maps constructed from layers of diagonal and Hadamard gates can be classically hard to simulate, providing a potential path toward quantum advantage in kernel-based classification tasks.

For image-specific applications, Henderson et al. [2020] introduced "quanvolutional" layers, quantum circuit analogues of classical convolutional filters. Small parameterised quantum circuits process local image patches, with pixel values encoded via rotation gates, and the measurement outcomes serve as extracted features fed into classical dense layers for classification. Demonstrated on MNIST, the approach showed that quantum circuits can produce useful nonlinear feature transformations with compact parameterisation.

Cong et al. [2019] proposed quantum convolutional neural networks (QCNNs) using $O(\log N)$ variational parameters for N-qubit inputs. The architecture draws on multi-scale entanglement renormalisation ansatz (MERA) and quantum error correction concepts, applying alternating convolutional and pooling layers that progressively reduce the system size. While demonstrated for quantum phase recognition rather than classical image tasks, the QCNN architecture has directly inspired subsequent work on quantum image processing.

Quantum autoencoders provide another building block. Romero et al. [2017] introduced parameterised quantum circuits trained to compress high-dimensional quantum states into fewer qubits while preserving the essential information. Training minimises a fidelity-based cost function evaluated via SWAP tests. The approach was demonstrated on compression of Hubbard model ground states and molecular Hamiltonians, and could be applied to compress quantum imaging data before classical post-processing.

6.3 Integration Challenges and Opportunities

The practical integration of quantum computational elements into imaging pipelines faces several concrete challenges.

Hardware limitations. Current quantum devices operate in the noisy intermediate-scale quantum (NISQ) regime [Preskill, 2018], with approximately 50 to a few hundred noisy qubits and no full error correction. Gate error rates of 0.1–1% limit useful circuit depth to roughly 10–50 layers before noise dominates the output. Short coherence times (approximately $100\,\mu$s for superconducting qubits) further constrain total computation time per circuit execution. These limitations mean that current quantum image processing demonstrations use heavily downsampled images (typically 4×4 or 8×8 pixels from MNIST) and binary classification tasks.

Trainability. McClean et al. [2018] identified the barren plateau problem: for wide classes of parameterised quantum circuits, gradient variance vanishes exponentially with qubit count, making gradient-based training impractical at scale. This is a fundamental scalability concern for quantum imaging models that require large circuits. However, Pesah et al. [2021] proved that QCNNs specifically avoid barren plateaus, with gradient variance vanishing at most polynomially. This makes QCNNs one of the few architectures with both favourable trainability properties and structural relevance to image processing, though a tension remains: circuits that provably avoid barren plateaus may in some cases be efficiently simulable classically.

Data loading. Encoding classical image data into quantum states presents a bottleneck. Amplitude encoding, which maps N pixel values to $\log_2 N$ qubits, requires circuit depth that scales exponentially with qubit count, making it infeasible for practical image sizes. Angle encoding, which uses one qubit per feature with rotation gates, avoids the depth problem but requires as many qubits as there are features, necessitating heavy classical preprocessing (PCA, classical autoencoders) to reduce dimensionality before quantum processing. This preprocessing overhead limits the potential quantum advantage.

Current experimental status. Hardware demonstrations of quantum image processing remain limited. Most experiments use IBM Quantum, Rigetti, or IonQ devices with classical preprocessing to reduce image dimensionality before quantum feature extraction, followed by classical classification layers. Results are typically competitive with classical models only on reduced, simplified problems. Scaling these approaches to practical imaging resolutions will require substantial advances in both quantum hardware and algorithm design.

7. Discussion and Open Problems

Several open problems cut across the domains reviewed above.

7.1 Benchmarking and Reproducibility

A critical gap in the current literature is the absence of standardised benchmarks for deep learning-based quantum imaging reconstruction. Different studies employ different image sizes, datasets, noise models, sampling ratios, and evaluation metrics, precluding rigorous cross-method comparison.

To address this, the community would benefit from adopting a fixed set of benchmark conditions: standard image sizes (28×28, 64×64, and 128×128 pixels), standardised noise models (Poisson noise at specific mean photon counts per pixel, e.g., 0.1, 1, and 10), common datasets (MNIST as a baseline, with CIFAR-10 or CelebA for evaluating performance on more complex natural images), and unified metrics (PSNR, SSIM, and perceptual quality measures such as LPIPS). The ghost imaging community would particularly benefit from shared experimental datasets with calibrated noise parameters and known ground truth, since most current work relies on individually simulated data with varying and often under-documented noise assumptions.

7.2 Scalability

Most reviewed deep learning methods operate on small images (28×28 to 64×64 pixels). Scaling to the high-resolution images required for practical applications introduces challenges in both network architecture design and training data generation. The computational cost of simulating quantum imaging systems at scale further compounds this limitation.

Several techniques from classical deep learning could help address the resolution gap. Progressive training, where the network is first trained at low resolution and then fine-tuned at progressively higher resolutions, can reduce training cost and improve convergence. Patch-based inference, where overlapping image tiles are processed independently and stitched together, allows networks trained on small images to be applied to arbitrarily large inputs. Attention mechanisms can capture long-range spatial dependencies without the quadratic memory cost of processing the full image at once.

Deep unfolding architectures [Monga et al., 2021] may scale more gracefully than purely data-driven methods, because the physics-based structure constrains the parameter space and reduces the amount of training data required at each resolution. For hybrid quantum–classical approaches, the qubit bottleneck discussed in Section 6 imposes an additional scaling barrier not present in purely classical pipelines.

7.3 Quantum Advantage in the Presence of Deep Learning

A fundamental open question is whether quantum correlations provide a genuine advantage over classical correlations when both are combined with state-of-the-art deep learning reconstruction. If neural networks can compensate for the additional noise in classical imaging, the practical quantum advantage may be smaller than previously assumed. The result of Cai et al. [2023], showing that deep learning can surpass the standard quantum limit using only classical light, underscores the importance of this question.

Systematic studies are needed that compare quantum and classical imaging under identical deep learning reconstruction pipelines, controlling for photon budget, detector efficiency, and image complexity. Such studies would clarify whether the quantum advantage lies primarily in the raw measurement quality, in the statistical structure of the noise (which a network might learn to exploit), or in some combination of both.

7.4 Hardware–Software Co-Design

The end-to-end optimisation paradigm demonstrated by Wang et al. [2019] for ghost imaging suggests a broader opportunity for hardware–software co-design, where the optical system and reconstruction algorithm are jointly optimised.

The concept of differentiable optics, where optical system parameters are treated as trainable variables in end-to-end gradient-based optimisation, has gained traction in classical computational imaging. For quantum imaging systems, this would involve jointly optimising SPDC source parameters (pump wavelength, crystal orientation, phase-matching conditions), spatial light modulator patterns, and the reconstruction network within a single differentiable pipeline. While the non-differentiable nature of photon detection events presents a technical challenge (requiring reparameterisation tricks or score-function gradient estimators), the potential for measurement reductions beyond what either component can achieve alone motivates further investigation.

Programmable photonic circuits present a particularly interesting platform for such co-design, since they can potentially serve as both the measurement apparatus and a component of the reconstruction computation, processing quantum optical states directly without classical conversion.

8. Conclusion

This survey has reviewed the field of deep learning-based reconstruction for quantum imaging systems, covering ghost imaging, compressed sensing with quantum correlations, sub-shot-noise imaging, and hybrid classical–quantum architectures.

Our principal findings are as follows. First, deep learning enables measurement reductions of one to two orders of magnitude in ghost imaging reconstruction, with end-to-end learned approaches providing the largest gains through joint optimisation of measurement and reconstruction. Second, the field is converging towards U-Net and GAN-based architectures, mirroring trends in classical computational imaging. Third, deep unfolding methods, which embed the physics of the forward model into the network structure, have recently been applied to ghost imaging and offer advantages in interpretability and data efficiency over purely data-driven approaches. Fourth, deep learning denoising is beginning to be applied to quantum-correlated imaging systems, with initial results suggesting that learned denoisers can complement or even substitute for quantum noise reduction in some settings.

Research Gaps

Several gaps in the current literature limit progress and should be prioritised:

(i) No standardised benchmarks or shared datasets exist for quantum imaging reconstruction, making rigorous cross-method comparison difficult.

(ii) Most methods have been tested only on small images (28×28 to 64×64 pixels); scalability to practical imaging resolutions is undemonstrated.

(iii) No controlled studies have compared quantum and classical correlations under identical deep learning reconstruction pipelines, leaving the practical quantum advantage unquantified.

(iv) Deep unfolding, which has proved effective in classical compressed sensing, has only recently been applied to ghost imaging and has not yet been explored for other quantum imaging modalities such as sub-shot-noise imaging.

(v) Hybrid quantum–classical architectures remain constrained by NISQ hardware limitations, data loading bottlenecks, and trainability issues such as barren plateaus.

Recommendations

Based on the gaps identified, we offer the following recommendations for the research community:

- Develop shared benchmark datasets incorporating both simulated and experimental quantum imaging data with calibrated noise parameters and known ground truth.
- Conduct systematic studies comparing quantum and classical imaging under identical deep learning pipelines, controlling for photon budget and detector characteristics, to quantify the practical value of quantum correlations.
- Explore deep unfolding approaches for quantum-specific measurement models beyond ghost imaging, particularly for sub-shot-noise imaging where the noise statistics differ from standard Poisson assumptions.
- As quantum hardware matures, investigate hybrid architectures that process quantum imaging data natively, without intermediate classical conversion, to preserve quantum information throughout the reconstruction pipeline.

Quantum imaging and deep learning is an active research area with scope for both fundamental and applied contributions. As quantum optical hardware improves and deep learning methods develop further, combining the two should yield imaging capabilities beyond what either can achieve on its own.

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